

Recording a route without satellites

How accurate a phone’s own sensors are, measured against GPS

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Abstract

When GPS is lost — in a tunnel, an underground car park, or an aircraft — a phone can still draw the route it travels. It reads *speed* from how the vehicle shakes and *direction* from its gyroscope and compass, and it never uses GPS after the first point. This paper measures the newest version of that system on a motorcycle, in a car, on foot and on a plane, by running it over every earlier recording and comparing the result with GPS recorded at the same time, used only as the answer key. Direction works in all four: it was within 30° of GPS 73% of the time on a motorcycle, 80% in a car, 93% on foot and 99% on a flight. Over 74 journeys (20 hours), distance came out 10% short on a motorcycle and within 1% in a car. Walking came out 4% long. In the air the app measures the takeoff and holds the speed it reached, which counted 61% of a flight’s distance. On a motorcycle, fast riding reads slow and slow riding reads fast. A phone with few examples of its own uses a small built-in network instead; on recordings it had never seen, it was as accurate as a phone with a full store.

1 What the system does

A route is a series of short steps. Each step needs a length and a direction. With GPS both come from satellites; here both come from the phone’s motion sensors.

Speed from vibration. A moving vehicle shakes, and it shakes differently at different speeds [1]. (Adding up acceleration instead does not work on a phone: its small sensor errors grow into kilometres within minutes [2].) Every second, the phone takes the last five seconds of up-and-down acceleration, measured 50 times a second, and turns it into a short “fingerprint” of eleven numbers — how much energy the shaking has at each of nine frequency ranges, plus two measures of its overall strength. Whenever GPS is available, the phone stores fingerprints together with the speed GPS measured at that moment; it keeps up to 4000 of them. Without GPS, it looks up the 12 stored fingerprints closest to the current one and averages their speeds:

$$\hat{v} = \frac{\sum_{j=1}^{12} v_j / d_j^2}{\sum_{j=1}^{12} 1 / d_j^2} \quad (1)$$

where v_j is a stored speed and d_j how far its fingerprint is from the current one. If even the closest stored fingerprint is far away, the phone does not guess. Averaging tends to pull every answer toward the middle, so a correction learned from the stored examples stretches the answers back out.

A starting model. The stored examples come from the phone’s own trips, so a new phone has none. Until it has 3000, it uses a small network instead: the same eleven-number fingerprint goes in, a speed comes out. The network was trained on recorded trips and ships with the app, so it needs nothing from the current trip. Like the store, it declines a fingerprint unlike anything it was trained on.

Speed in the air. A smooth cabin reads as standing still, so in the air the app measures the takeoff instead. From the last moment the phone was still, it follows the phone’s turning with the gyroscope and adds up the push the accelerometer feels, for up to two minutes; then it holds the speed it reached.

Direction from turns. The gyroscope measures every turn precisely but does not know north; the compass knows north but measures where the *phone* points, not where the vehicle is going — a known problem for phones carried in different ways [3]. In a pocket those differ by some angle β , which depends on how the phone happens to sit. The phone learns β from the vehicle’s own turns. In a turn the vehicle is pushed toward the inside of the curve, so the sideways push the phone feels, compared with which way the gyroscope says it is turning, shows where “forward” is. Written as complex numbers,

$$\hat{\beta} = \arg \left(\frac{\sum_R z_t}{\sum_R |v_t \omega_t|} + \frac{\sum_L z_t}{\sum_L |v_t \omega_t|} \right), \quad z_t = \frac{a_t}{|a_t|} i \overline{v_t \omega_t} \quad (2)$$

where a_t is the sideways and forward acceleration measured relative to the compass, v_t the speed, ω_t the turn rate, and R and L the seconds turning right and left. Braking into a corner and pulling away out of it lean the answer one way in right turns and the other way in left ones, so the two kinds are averaged apart. Only seconds above 14 km/h count, and only when the turn explains at least a tenth of the push: otherwise the push is braking or throttle, and a reversing car’s points backwards. The phone makes a first estimate after 30 seconds of riding and keeps improving it; when it improves, the part of the ride already drawn is redrawn with the better value.

Walking is handled differently: over four seconds, the back-and-forth of the steps lines up with the direction of walking, and each push-off is sharper than each landing, which says which end of that line is forward.

The compass reading itself. The phone’s built-in heading is unreliable in one common pocket position: when the phone sits head-down at an angle, it scattered by about 49° a minute even though the phone was not moving in the pocket. So the system computes its own heading from the phone’s orientation, using whichever edge of the phone is closest to level. That heading scattered by about 1° a minute in the same position.

In a car the heading also drifts with the gyroscope’s own small bias: 42° in 25 minutes on one drive, the phone fixed. A stopped car does not turn, so the phone measures the heading’s turning while stopped and still, as a rate (at most 3° a minute), and takes it off.

Only the first point uses GPS. After that, each step is laid down from the previous point, along the current direction, for the distance the speed covers in that second.

2 How it was tested

GPS was recorded alongside every test and never used by the system; it serves only as the answer key. So that every result describes the newest version, the newest method was run again over all earlier recordings, which were made with the phone’s raw sensor readings saved — its acceleration, rotation and orientation angles, fifty times a second:

- **Speed.** The app’s speed logic was run again on the recorded acceleration: the same fingerprint, the same 12-neighbour average, the same refusals and correction, and the same rules for a stopped vehicle, for a phone being handled, and for holding the last answer. Its memory holds 4000 examples drawn at random from every *other* journey, so that no journey was ever measured with its own GPS in the memory. On twelve recent journeys the re-run came out typically 4% shorter than what the app itself had drawn.
- **Direction.** The direction method, including the drift correction and the redrawing, was run again on the recorded rotation, orientation and acceleration, with the re-run speed.
- **Starting network.** Each recording was measured by a network trained without it (the recordings were split into eight groups), and the store was tested holding only 60 to 4000 examples.
- **Walking.** The app’s step detector was run again on the recorded acceleration, with the stride the phone had learned. The phone’s own step counter, which the app also consults, was not recorded; its share is taken from each journey’s first walk, where it added about 2% on average.

Three inputs are taken from the recordings as they were: whether each second was walking or riding (which older versions decided partly with GPS help the current one no longer uses), Apple’s motion classifier, and the car-park ramp flag. Cabin pressure was not recorded, so on the flight the moment the app would call it airborne was taken from the same integration’s climb rate.

Every result is split by what was moving. A motorcycle and a car were told apart from the sensors themselves: in a turn a motorcycle leans and the phone leans with it, while a car stays level and the phone is pushed sideways. This agreed with every journey whose vehicle was known. Walking is the time the app was counting steps. That gives 39 motorcycle journeys (11 hours, the phone in a trouser pocket) and 35 car journeys (8.6 hours, the phone in a pocket or lying flat) for speed and distance, 68 routes, 65 recordings for direction, 45 straight half-minutes and seven longer walks on foot, and one flight. Four car journeys went into, started in or ended in underground car parks, where GPS kept reporting positions it did not have; those stretches, found from the barometer, are left out (Section 4). Half-minutes that mix walking and riding, or in which the app counted steps while GPS showed riding speed, are left out.

In a pocket GPS is itself only accurate to 10–40 m, so it was used over stretches long enough for that not to matter:

- **Speed and distance** are compared over half-minute stretches between GPS fixes that are better than 40 m: the straight distance GPS moved, against the distance the system added. Walking moves so little in half a minute that only straight stretches between fixes better than 25 m are used.
- **Direction** is compared every second: the bearing from the GPS position 15 s earlier to the one 15 s later, against the system's heading over the same half-minute. For the plane, which covers far more ground, the stretch is ten seconds.
- **Whole routes** are compared by finding the rotation and the size change that best lay the drawn route over the GPS track, pinned at the shared starting point. The rotation is how far the whole route is turned; the size change is how much too big or too small it is.

3 Results

Every graph is split by what was moving: a motorcycle, a car, a person walking, or a plane.

3.1 Speed

- **Motorcycle** (818 half-minutes). The app squeezes the range: 10–20 km/h reads about 19, 20–40 km/h is close, and above 60 km/h it reads about 41 where GPS measured 66. The squeeze comes from averaging twelve examples: when a fingerprint could belong to several speeds, the average lands in the middle. The average error is 9 km/h.
- **Car** (733 half-minutes). Much closer: from 10 to 60 km/h the middle reading is within about 2 km/h of GPS. Above that it reads low (62 where GPS measured 69, and 71 where it measured 87). The average error is 7.3 km/h.
- **Walking** (45 straight half-minutes). About 5 km/h where GPS measured about 4.6, just after a ride as at any other time.
- **Plane** (one flight). While taxiing the app reads about 17 km/h where GPS measured 18. In the air it measures the takeoff — 265 km/h forty seconds into the roll, where GPS measured 266 — and holds the 317 km/h it reached; it does not see the aircraft keep speeding up to 680 km/h in the climb.

3.2 Distance

- **Motorcycle** (39 journeys, 176 km checked). The app added 158 km, 10% short: the distance lost at high speed outweighs what is gained at low speed. Of the 38 journeys with at least 1 km checked, the middle one was 5% short and 79% were within 20%; the worst were 65% short and 64% long.
- **Car** (35 journeys, 163 km checked). The app added 163 km, 0.3% short. Of the 31 journeys with at least 1 km checked, the middle one was 1.6% short and 65% were within 20%; the worst were 37% short and 67% long.

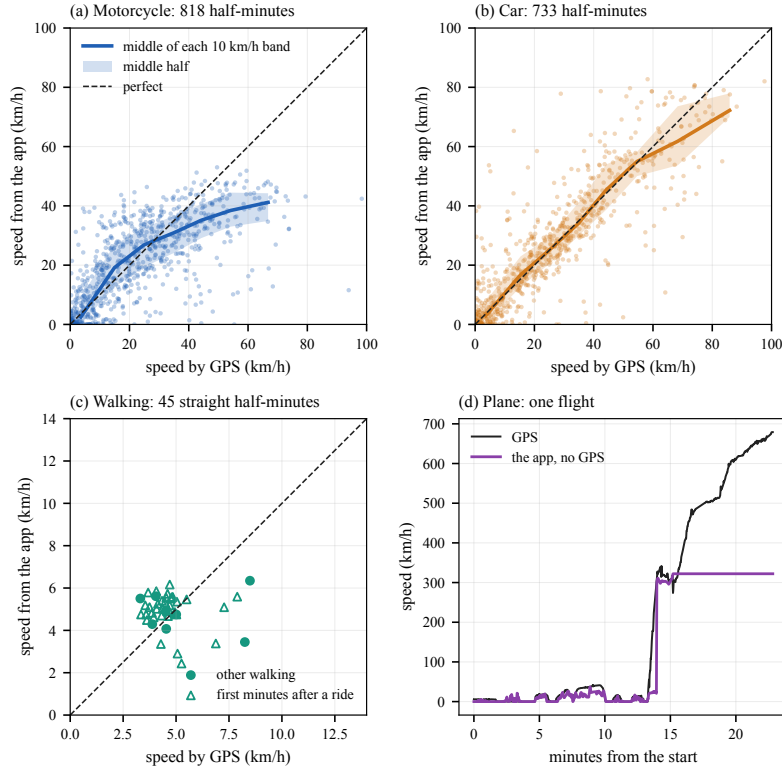


Figure 1: Speed from the app against speed from GPS. (a, b) Each dot is a half-minute of riding; the solid line is the middle of each 10 km/h band and the shading holds the middle half. On the dashed line the two agree. (c) Straight half-minutes of walking; triangles are the first minutes after a ride. (d) One flight, over time.

- **Walking.** Over straight stretches the app was 4% long: 7% long in the first minutes after a ride and 7% short at other times.
- **Plane.** The app counted 45 km of the 74 km flown (61%).

Motorcycle and car together, the app added 321 km where GPS measured 340 km, 5.5% short.

3.3 Direction

Direction holds up in every case (Figure 3):

- **Motorcycle** (40 recordings): within 30° 73% of the time; middle error 16° .
- **Car** (25 recordings): within 30° 80% of the time; middle error 17° .
- **Walking** (seven walks with the phone in a pocket): within 30° 93% of the time; middle error 10° .
- **Plane** (one flight): within 30° 99% of the time; middle error 6° .

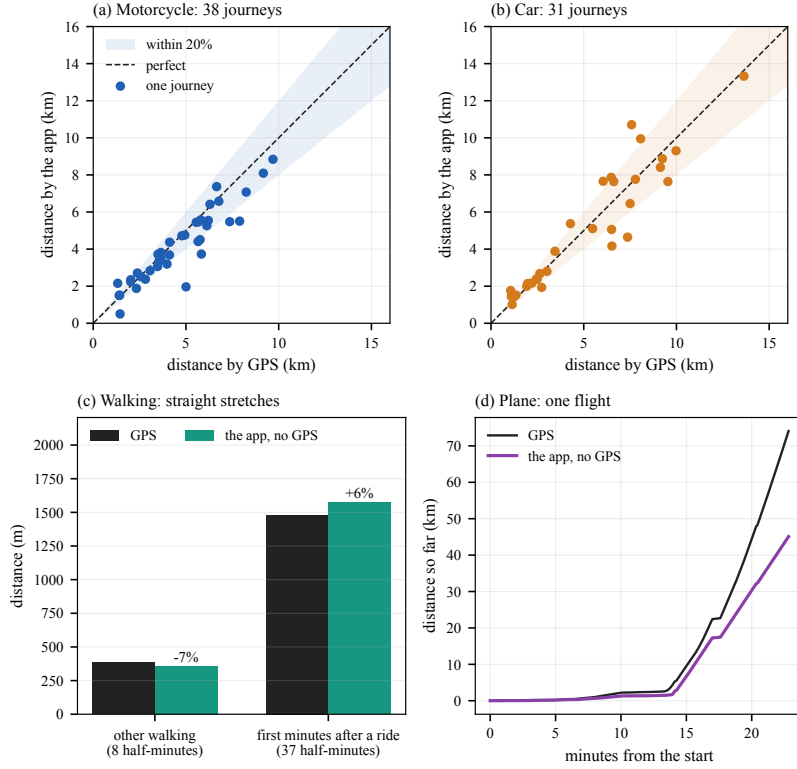


Figure 2: Distance by the app against GPS. (a, b) Each dot is one journey with at least 1 km that GPS could check; the shading is within 20%. (c) Straight stretches of walking, added up. (d) Distance so far on one flight.

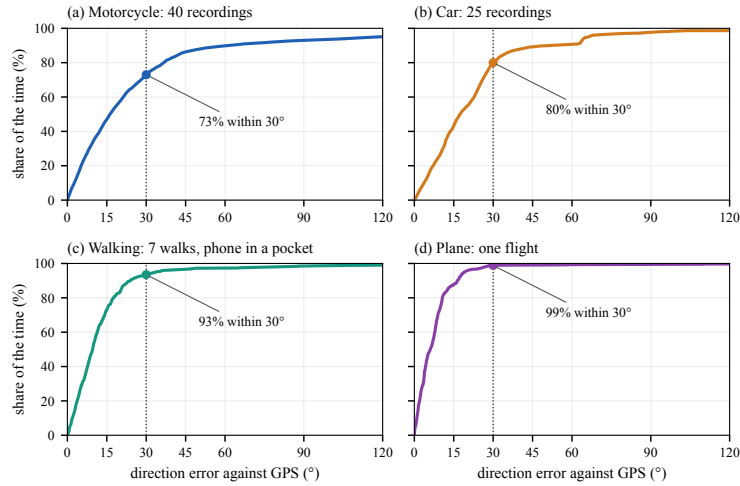


Figure 3: How often the direction is within a given angle of GPS, compared every second with a half-minute of GPS track (ten seconds for the plane).

3.4 Whole routes

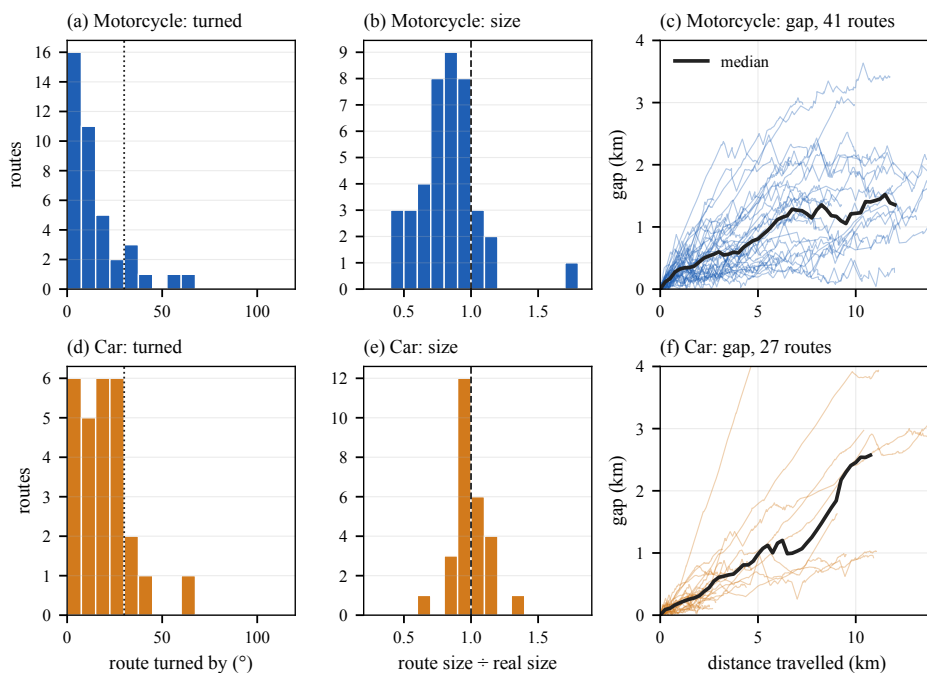


Figure 4: Routes drawn by the app, compared with GPS. (a, d) How far each whole route is turned from the GPS track; the dotted line is 30° . (b, e) Its size relative to the GPS track. (c, f) How far the drawn position is from the GPS position as the journey goes on: one thin line per route, and the median.

Table 1: Summary: the newest version against GPS.

	Distance	Direction within 30°	Whole route turned
Motorcycle	10% short; 79% of journeys in 20%	73% of the time	median 9°
Car	0.3% short; 65% of journeys in 20%	80% of the time	median 17°
Walking	4% long	93% of the time	—
Plane	45 of 74 km	99% of the time	—

Figure 4 and Table 1 put speed and direction together.

- **Motorcycle** (41 routes, 351 km, many with a walk at one end). The median route is turned 9° and 83% are within 30° . Routes come out small — a median 0.84 of the real size — partly because distance is short, and partly because small direction errors make the path fold. The drawn position drifts from GPS by a median of 17% of the distance travelled.
- **Car** (27 routes, 146 km). The median route is turned 17° and 85% are within 30° ; the size is right (median 0.98). The drift is 19% of the distance travelled.
- **Walking**. No route was more than half walking.

- **Plane.** Not compared as a route: its direction is in Section 3.5 and its distance in Figure 2d.

Four routes are turned more than 45° . Three of them are short journeys, under 3 km, with too few turns to learn the pocket angle well.

3.5 The flight, second by second

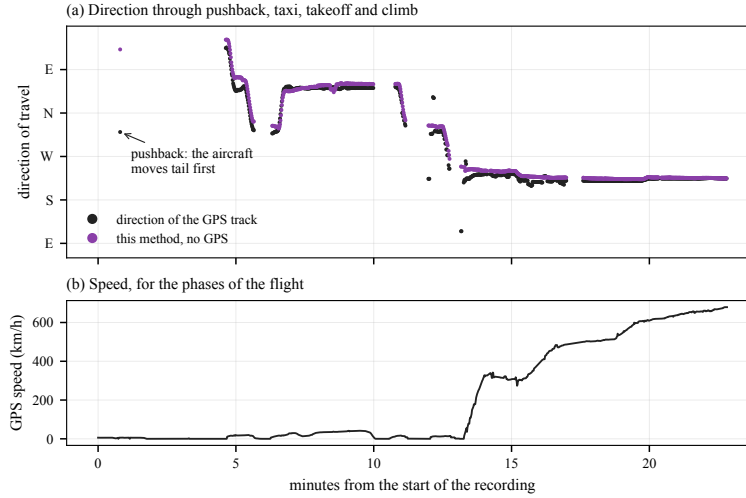


Figure 5: One flight, from the gate to the climb. (a) The direction of travel from the GPS track (black) and from this method, which uses no GPS (purple). (b) The aircraft’s speed from GPS, to show where it was taxiing, taking off and climbing.

Figure 5 follows the flight with the same direction method as on the road. While taxiing, the middle error was 9° ; in the air it was 4° . Ten seconds were more than 30° off: the pushback, when the aircraft moves tail first while the method gives the way the nose points, and nine while it was stopped or crawling before the runway, where the GPS track has no clear direction. In the climb the error stays small, about 1° by the end.

3.6 A new phone

A new phone has no examples of its own, so it starts with the built-in network. Figure 6 puts it next to the main model with a full store, kind of travel by kind of travel:

- **Motorcycle.** The network misses by 9.1 km/h on average and the main model by 9.0. The network counted 91.5% of the distance and the main model 90.0%; 79% of journeys came within 20% with either.
- **Car.** The network misses by 6.7 km/h and the main model by 7.3; they counted 101.7% and 99.7% of the distance, with 74% and 65% of journeys within 20%.
- **Walking.** The same, 103.7%: walking is counted by steps, not by either speed model.
- **Plane.** The same, 61%: in the air both hand over to the measured takeoff.

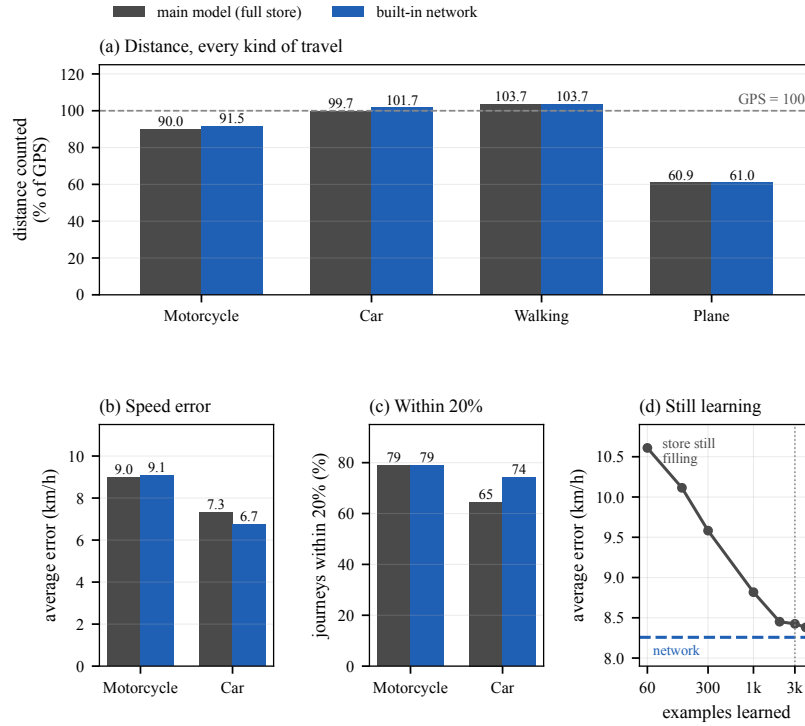


Figure 6: The built-in network (blue) next to the main model with a full store (grey), measured on the same journeys; the network only on recordings it had never seen. (a) Share of the GPS distance counted. (b) Average speed error. (c) Journeys within 20% of GPS. (d) A store that is still filling up, against the network; the dotted line is where the app hands over.

A store that is still filling does worse (Figure 6d): with 60 examples the average error was 10.6 km/h, with 1000 it was 8.8, and it draws level with the network (8.3) at about 3000. That is where the app hands over from the network to the store, which then keeps adapting to the phone’s own vehicle and pocket.

4 Where it goes wrong

- **On a motorcycle, fast riding reads slow and slow riding reads fast.** Above 60 km/h the speed is under two-thirds of the true value (Figure 1a). Over a whole journey the two partly cancel, but a journey that is mostly fast comes out short. A car does not have this problem below 60 km/h.
- **The first minute of a ride.** Until the pocket angle is learned, the direction comes from the compass alone, which in a pocket can be far off. The redrawing fixes the saved route afterwards, but a very short ride may never learn the angle well.
- **A phone that is picked up.** A hand shakes the phone, and shaking is what the speed model reads: going down into a car park, a lifted phone read 88 km/h. A hand

can no longer raise the speed (re-run, that stretch reads about 6 km/h), but a phone picked up while slowing keeps its last speed until it is still: once 48 km/h for ten seconds after stopping. A phone that shifts in the pocket turns the rest of the ride by about as much as it moved.

- **Underground car parks.** GPS cannot check the app there: in four car journeys it kept claiming 10 m accuracy underground, its position frozen while parked and wandering on the ramps, so those stretches are left out. Parked underground for five and a half minutes, the app added 2 m. A car crawling up a concrete ramp shakes like one driving at 40 km/h, so distance is not counted while climbing and turning together; once that test fired on an ordinary road and lost about 115 m.
- **A magnet beside the phone.** By a car’s MagSafe charger the phone read up to 2,600 μT , fifty times Earth’s field, yet called its compass well calibrated. No compass works then: the iPhone’s own Compass drifted 49° with the gyroscope, as the app did.
- **Rides never recognised as driving.** Without GPS, the app knows it is in a vehicle from Apple’s motion classifier or its own speed reading. On one short pocket ride the classifier never said “driving” and the step counter took the engine for footsteps: 59% short.
- **Aircraft after the takeoff.** The app holds the speed the takeoff reached, so it misses the speed-up in the climb: it held 317 km/h while the aircraft reached 680, and counted 61% of the distance (Figure 1d). The phone’s own orientation filter takes a takeoff’s steady push for a tilt [4], which is why the raw accelerometer is used. Examples learned on earlier flights with GPS would take over, which one flight could not test.

5 What did not work

Measured against GPS on the same recordings, none of these did better:

- **Direction from speeding up and braking.** The push when accelerating in a straight line points forward, so it should give direction. It was 44° off; turns work far better.
- **Leaving the speed out of the turns** (weighting by the change in turn rate): 15–40° off on one drive, as braking into and out of corners added up.
- **Other direction ideas:** the axis the motorcycle leans about (25° off), the direction of the rider’s thigh (73° off), and the swing of the thigh for telling forward from backward when walking (wrong more often than right).
- **Other ways to tell riding from walking:** a steady push also fired while walking, and letting the speed overrule steps blocked 5% of real walking.

- **Following the rest of a flight.** Speed from turns: the phone’s own turning (0.4–1°/s) hid the gentle turns. Integrating the whole climb: one setting only.
- **A better starting network.** Adding the last ten seconds helped one fingerprint at a time (6.3 against 8.0 km/h) but not over journeys.

6 Lessons

- **The phone’s own definitions caused more error than the models.** Its built-in heading becomes unreliable in a common pocket position and drifts slowly in a car, and its “user acceleration” points the opposite way to the vehicle’s acceleration. Each of these, once found, mattered more than any change to the estimators.
- **GPS can hide in decisions as well as in distance.** The no-GPS mode never drew a metre from GPS, yet it still used GPS to decide whether the phone was riding or walking, and to set the first heading. Only reading every place the mode touched GPS found them; both are gone.
- **A re-run must do everything the app does,** stops and a phone in the hand included; with those rules in, it comes within about 4% of what the app drew.
- **Check what a counter stores while nothing reads it.** The step detector counted road bumps as steps all ride and only a walk emptied it, so the next walk came out twice its length.

Limits of this evaluation

All recordings come from one person and one iPhone, mostly in one city, with only one flight; other people, phones and vehicles may behave differently, the starting network most of all. Most results re-run the newest method on earlier recordings rather than coming from live use, and GPS in a pocket is only accurate to tens of metres, so small errors cannot be measured reliably.

References

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- [4] M. B. Del Rosario, N. H. Lovell and S. J. Redmond. Quaternion-based complementary filter for attitude determination of a smartphone. *IEEE Sensors Journal*, 16(15):6008–6017, 2016.